Decision Making with Sentiment Analysis in R

From Social Media Noise to Managerial Action

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Research Question

How can customer tweets be transformed into decision-relevant insights for airline management?

What is Sentiment Analysis?

Sentiment analysis is a text mining method used to identify opinions, emotions, and attitudes in written language.

Positive

Neutral

Negative

Why It Matters for Decision-Making

Companies receive large amounts of customer feedback.

Managers need fast and structured insights.

Sentiment analysis helps prioritize business actions.

Different Sentiment Analysis Methods

Method	Basic idea	Why not used / selected?
Lexicon-based	Uses predefined sentiment dictionaries	Selected because it is transparent and easy to reproduce in RStudio
Machine learning	Trains a model on labeled text data	Requires training/testing setup
Transformer-based	Uses context-aware models such as BERT	More complex for a short live demo
Hybrid	Combines dictionaries and machine learning	More advanced setup

In this project, I use a lexicon-based approach because it is transparent, reproducible, and suitable for a short RStudio demonstration.

From Text to Managerial Action

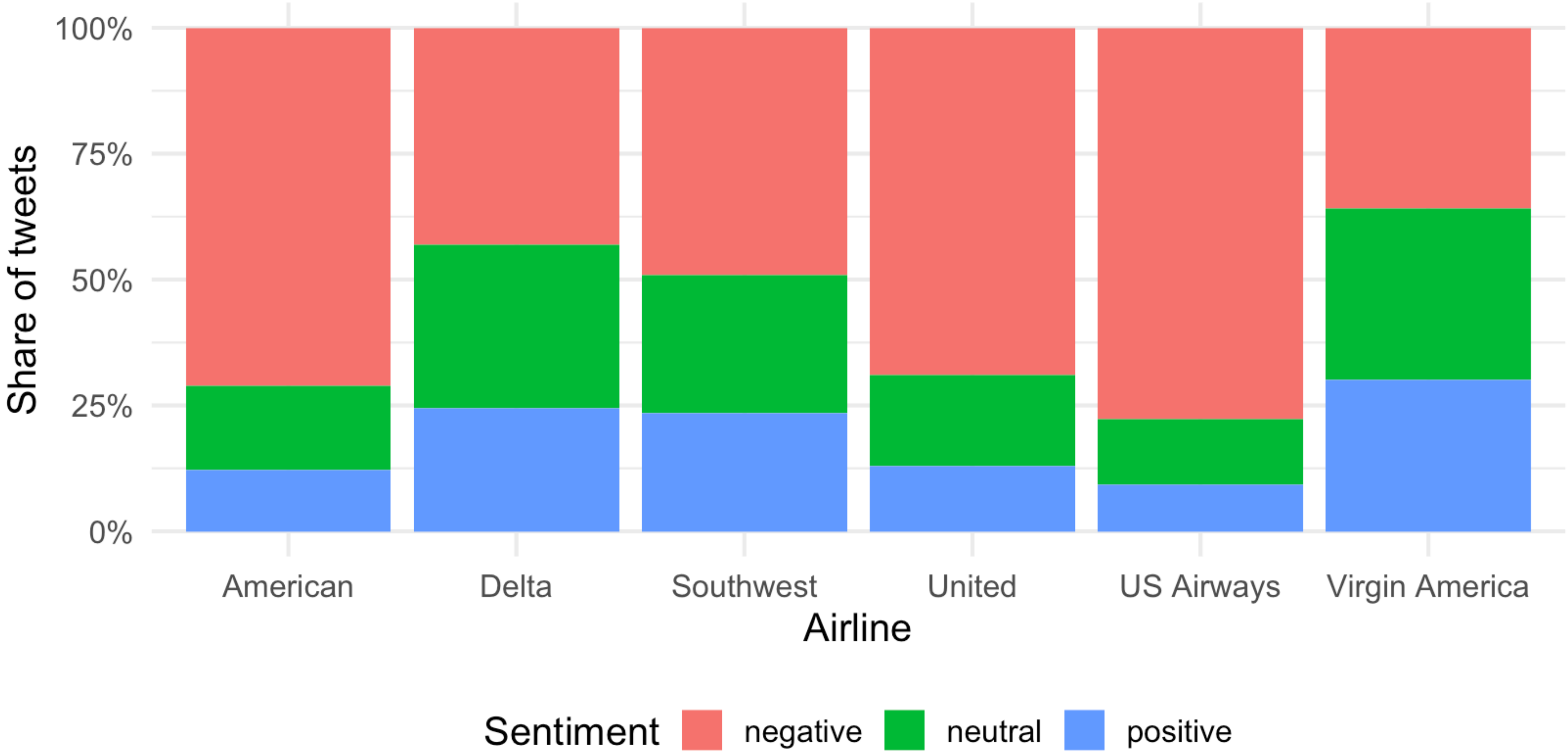
1	2	3	4	5
Collect text	Clean data	Measure sentiment	Find problems	Recommend action

Case Study: Airline Tweets

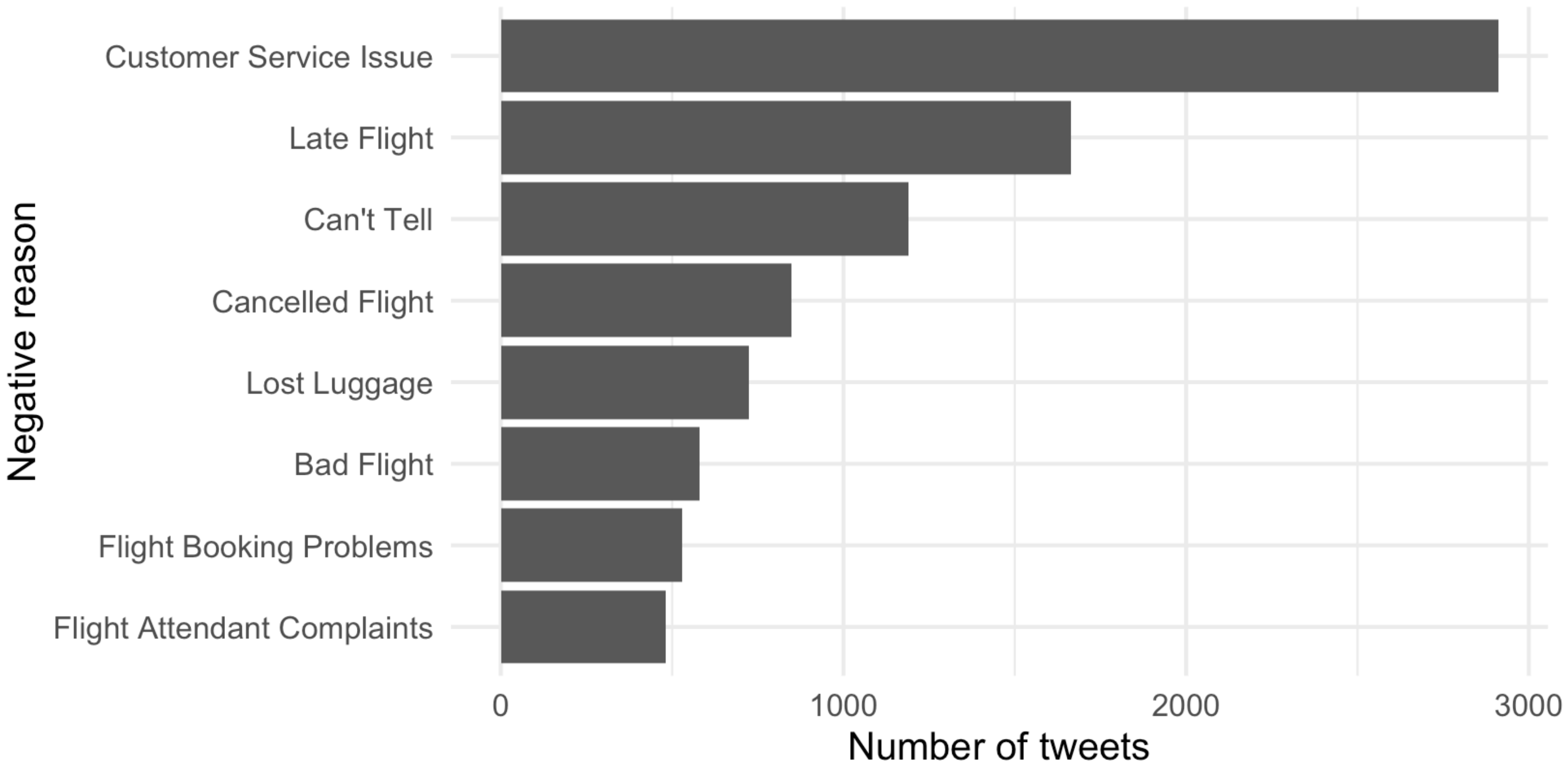
The example uses **14,640 airline-related tweets** classified as negative, neutral, or positive.

Sentiment	Tweets	Share
negative	9178	62.7%
neutral	3099	21.2%
positive	2363	16.1%

Sentiment by Airline



Main Negative Reasons



R Workflow: Tokenization

```
tweets_words <- tweets |>
  mutate(tweet_id = row_number()) |>
  select(tweet_id, airline, airline_sentiment, text) |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = "word")
```

Example
The flight was delayed and the service was terrible.

After tokenization:

flight delayed service terrible

Lexicon-Based Sentiment Analysis

```r
word_sentiments <- tweets_words |>
  inner_join(get_sentiments("bing"), by = "word")
```

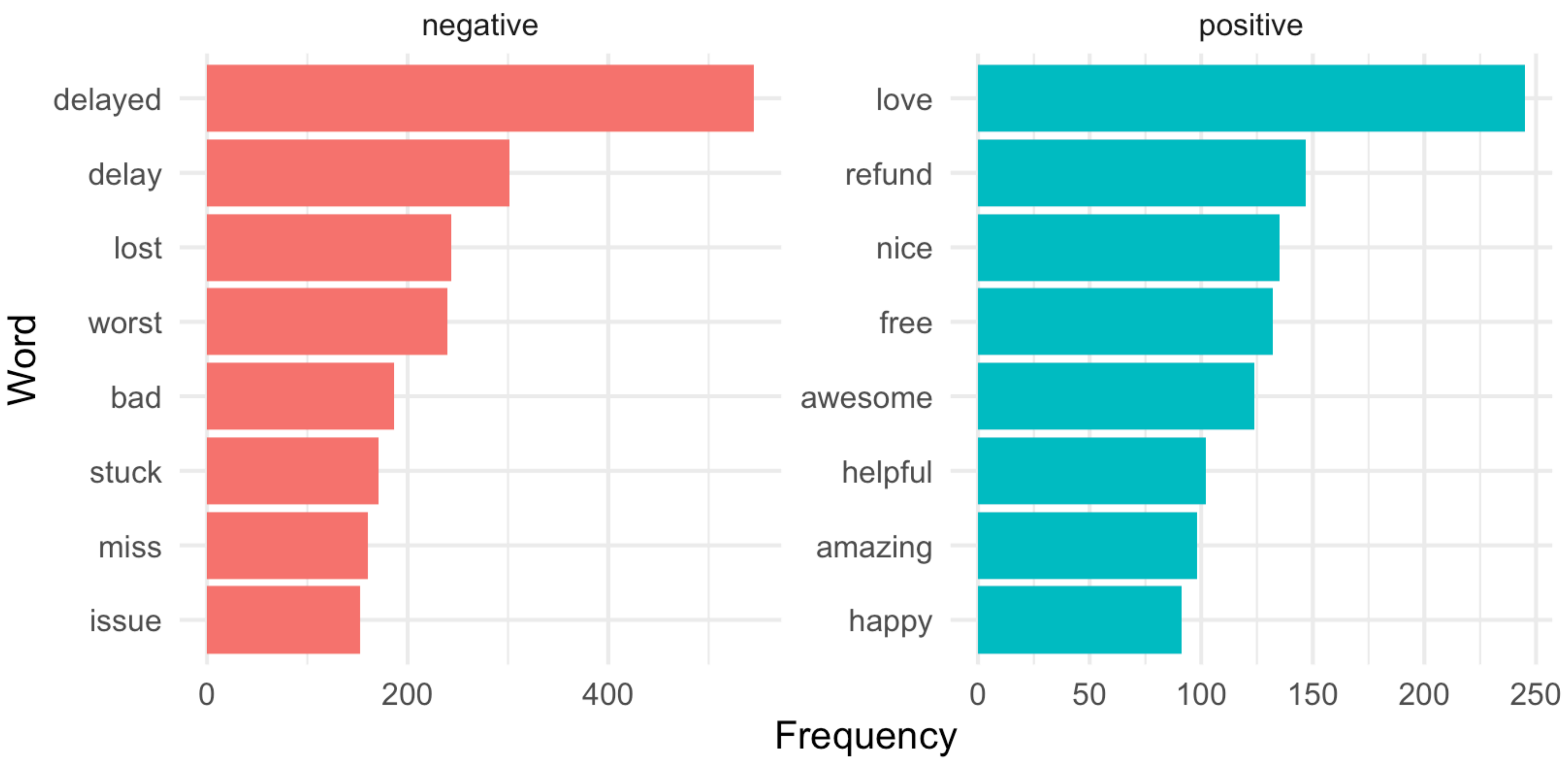
Example word matching with Bing lexicon

The flight was delayed and terrible, but the staff was helpful and comfortable.

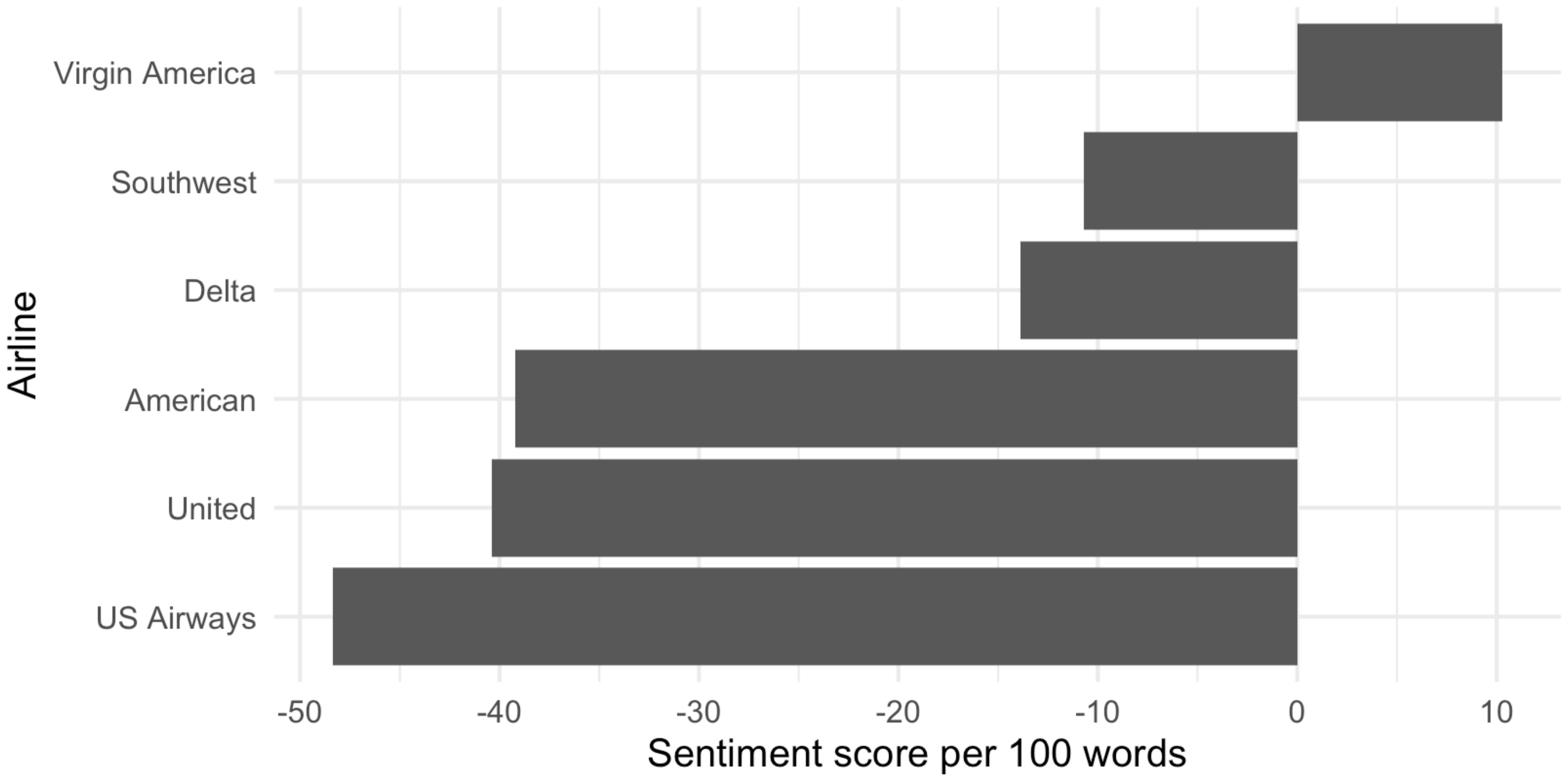
terrible → negative delayed → negative helpful → positive comfortable → positive

Words that are not found in the dictionary are not labelled as neutral; they are excluded from the lexicon-based count.

Frequent Sentiment Words



Sentiment Score by Airline



From Analysis to Decisions

Problem	Tweets	Managerial action
Customer Service Issue	2910	Improve support capacity and service recovery.
Late Flight	1665	Improve delay communication and punctuality.
Can’t Tell	1190	Investigate this category further.
Cancelled Flight	847	Strengthen rebooking and compensation.
Lost Luggage	724	Improve baggage tracking.

Managerial Interpretation

The real value is not only knowing whether customers are happy or unhappy.

The real value is knowing which operational problem should be solved first.

Strengths and Limitations

Strengths

Fast analysis
Reproducible workflow
Useful for monitoring

Limitations

Sarcasm is difficult
Context can be missed
Human interpretation is needed

In-Class Exercise

Students analyze three short customer comments in R and recommend one managerial action.

```
customer_comments <- tibble(
  text = c(
    "The flight was delayed and the service was terrible",
    "The staff was helpful and the flight was comfortable",
    "My luggage was lost and nobody helped me"
  )
)

customer_comments |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = "word") |>
  inner_join(get_sentiments("bing"), by = "word") |>
  count(sentiment)

# Show which words were matched as positive or negative
customer_comments |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = "word") |>
  inner_join(get_sentiments("bing"), by = "word") |>
  select(word, sentiment) |>
  arrange(sentiment, word)
```

Exercise Question

Question:
Which managerial action would be most suitable based on the negative words in the comments?

A) Reduce the number of customer support channels
B) Improve delay communication and luggage recovery processes
C) Ignore negative comments because they are unstructured data
D) Only focus on positive words

Conclusion

Sentiment analysis turns unstructured customer feedback into practical decision support.

For managers, the key question is not only: **“What is the sentiment?”**
It is: **“What decision should this sentiment trigger?”**

References

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Pang, B., & Lee, L. (2008). Opinion mining and sentiment analysis. *Foundations and Trends in Information Retrieval*, *2*(1–2), 1–135.

Silge, J., & Robinson, D. (2017). *Text mining with R: A tidy approach*. O’Reilly Media.

Twitter US Airline Sentiment dataset. Airline-related tweets used for sentiment and negative reason analysis.

Thank You

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